

Supplementary Information

A Physiology-Aligned State-Space Decomposition Framework for Camera-Derived Respiratory Motion

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Supplementary Methods

Label-free reliability law

The reliability evidence used by PARH-OSSM is computed without using the respiratory reference labels. Each candidate observation contributes label-free features describing rate agreement, preprocessing invariance, event timing, bridge timing, rate/phase support, morphology support, ROI observability, harmonic ambiguity, nuisance evidence, and abstention pressure. Reference-derived or oracle columns are removed before materialization of the evaluated runs.

All reliability features are mapped to $[0, 1]$ by clipping. Let G denote group support, M macro timing, B bridge timing, P rate/phase support, O_{roi} ROI observability, O_{prep} preprocessing invariance, T timing reliability, E event timing, U nuisance evidence, H harmonic-risk score, H_g harmonic-guard score, R_m morphology reliability, M_s morphology support, R generic reliability, and A_b abstention pressure. The component-law implementation uses the fixed combinations

$$A = \text{clip}_{[0,1]}(0.35G + 0.25M + 0.20B + 0.20P), \quad O = \text{clip}_{[0,1]}\{\sqrt{O_{\text{roi}}O_{\text{prep}}}\},$$

$$T^* = \text{clip}_{[0,1]}\{T(1 - 0.55H)\},$$

$$w_{h1} = \text{clip}_{[0,1]}\{T^*(0.55 + 0.25E + 0.20A)(0.65 + 0.35O)(1 - 0.45U)\},$$

$$w_{h2} = \text{clip}_{[0,1]}\{M_s(0.45 + 0.35H_g + 0.20G)(0.65 + 0.35O)(1 - 0.30U)\},$$

$$w_b = \text{clip}_{[0,1]}\{(0.45O_{\text{prep}} + 0.35O_{\text{roi}} + 0.20R)(1 - 0.35U)\},$$

$$N_m = \text{clip}_{[0,1]}\{\max(R_m - T, 0)\},$$

$$w_r = \text{clip}_{[0,1]}\{(0.55M_s + 0.25G + 0.20N_m)(0.70 + 0.30O)(1 - 0.55U)\},$$

and

$$A_c = \text{clip}_{[0,1]}\{\max(A_b, H(1 - T), U(1 - G), 0.75(1 - O))\}.$$

The oscillatory readout anchor and full-waveform readout weights are then

$$w_{z_{\text{osc}}} = \text{clip}_{[0,1]}\{w_{h1}(1 - 0.50A_c)\},$$

$$w_{z_{\text{full}}} = \text{clip}_{[0,1]}\{(0.35w_{h1} + 0.25w_{h2} + 0.20w_b + 0.20w_r)(1 - 0.35A_c)\}.$$

These weights are responsibilities for state and readout roles, not a ground-truth selector. They modulate observation trust and process-noise allocation; respiratory labels are used only for evaluation and post hoc oracle analyses.

Supplementary Tables

Supplementary Table S1. Dataset and evidence scope.

Dataset	Role	Label scope	Main use	Supported claim
COHFACE	Primary real benchmark	Respiration waveform and derived rate	Rate, aligned waveform, strict waveform, and cycle diagnostics	Supports clean-regime real waveform/rate claims.
MAHNOB-HCI	Hard real benchmark	Respiration-belt waveform and derived rate	Rate, aligned waveform, strict waveform, and observability taxonomy	Supports hard-regime boundary analysis; low observability is reported explicitly.
V4V	External real rate-only evidence	Frame-aligned RR labels; no raw respiratory waveform	External rate-scope check	Does not support waveform morphology claims.
SCAMPS	Synthetic diagnostic control	Synthetic breathing signal and video arrays	Controlled mechanism and distribution diagnostics	Not mixed with real-data performance claims.

This table defines the evidence boundary used throughout the manuscript. The real-data claims come from COHFACE and MAHNOB-HCI, whereas V4V and SCAMPS are retained only for rate-scope or diagnostic context. This separation prevents external or synthetic controls from being mixed into the real waveform benchmark.

Supplementary Table S2. Fixed observation-class map.

Class	Construction	Primary information	Nuisance risk	PARH-OSSM role
OF	Raw vertical optical-flow surrogate	Oscillatory motion and local phase/frequency	Medium	Helper-heavy rate evidence.
OF bridge	OF-derived displacement-compatible bridge	Bridged oscillatory displacement	Medium	Rate-oriented constructed cue.
DoF	Thresholded frame-difference count	Thresholded motion energy	High	Auxiliary nuisance-limited cue.
DoF bridge	Signed DoF motion-centroid trajectory	Coarse signed motion trajectory	High	Hard-regime signed-motion probe.
P1D linear	Linear one-dimensional profile shift	Fundamental displacement	Medium	Conservative displacement cue.
P1D quadratic	Quadratic one-dimensional profile shift	Harmonic morphology and asymmetry	Low	Waveform-primary harmonic cue.
P1D cubic	Cubic one-dimensional profile shift	Harmonic morphology and asymmetry	Low	Waveform-primary harmonic cue.
P1D consensus	Weighted consensus of P1D routes	Consensus displacement morphology	Low	Agreement and morphology-support cue.

This table is the nomenclature map for the fixed observation bank. The eight classes are not interchangeable preprocessing variants: they expose different motion cues, nuisance risks, and likely roles in the PARH-OSSM

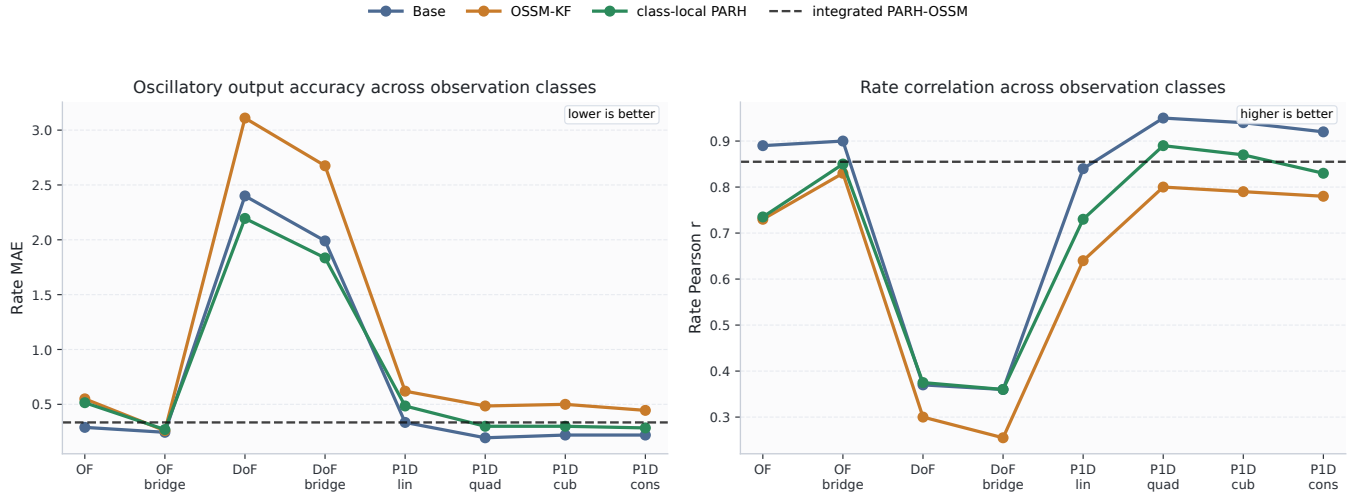
state/readout pipeline. The table therefore supports the main-text claim that observability must be analysed at the observation-class level.

Supplementary Table S3. Supplementary data files included with the release package.

File	Purpose
Strict waveform CSV	Strict zero-lag waveform metrics and span-normalized reconstruction diagnostics.
Cycle timing CSV	Cycle timing companion metrics for peak, trough, and peak-to-peak intervals.
Readout-alignment ablation CSV	COHFACE readout-alignment ablation supporting rate/waveform role separation.
Component-ablation evidence CSV	Component-level evidence and claim-boundary ledger.
Diagnostics CSV	Calibration diagnostics, including PARH NIS and Student- t weights.
Observation-class comparison CSV	Complete fixed-observation, OSSM-KF, and PARH-OSSM diagnostic comparison.
Paired statistical comparison CSV	Paired common-subset effect-size deltas used to interpret MAHNOB-HCI fairly.

This table links the numerical claims to the released audit artifacts. It distinguishes strict waveform metrics, observation-class comparisons, and paired common-subset statistics so that readers can reproduce the main claims without treating diagnostic-only files as hidden model-selection inputs.

Supplementary Figures



Supplementary Fig. S1. Complete eight-observation rate diagnostic plot. The figure compares direct observations, OSSM-KF, and observation-class-local PARH timing behavior across the fixed observation classes and is used only as diagnostic context, not as a post hoc best-source selector.

This figure expands the rate evidence beyond the three compact rows in the main text. It shows whether fixed observations, OSSM-KF, and PARH-style readouts fail for the same reason or for different observation-class reasons. As diagnostic context, it explains the observed boundary without defining a post hoc best-source policy.

This figure localizes where observation quality changes before the latent state model is applied. The preprocessing views should therefore be read as parallel evidence transformations, not as a single ordered cascade. Their main purpose is to show which cues survive simple transformations and which become nuisance-dominated.

This figure clarifies why OF and OF-bridge are kept as separate observation routes. Raw optical flow and the bridged displacement-compatible cue can differ in rate support, waveform topology, and failure mode, so collapsing them would hide a construction-level source of evidence.

This figure defines which explored components are supported by numeric performance evidence and which are retained only as diagnostics or construction probes. It guards against overclaiming by making the difference between promoted architecture and audit evidence explicit.

This figure explains the role of datasets that are informative but not part of the main waveform benchmark. V4V contributes external rate-label context and SCAMPS contributes synthetic diagnostic control, but neither supports the real-video waveform morphology claims made from COHFACE and MAHNOB-HCI.

This figure exposes the interface between candidate evidence and latent state roles. It helps readers check whether harmonic, baseline, and residual drives are being activated by plausible observation summaries rather than by an unreported oracle source.

This figure supports the dual-readout interpretation in the main text. It shows that confidence and agreement can differ between the oscillatory rate path and the full waveform path, which is why z_{osc} and z_{full} are not evaluated as the same target.

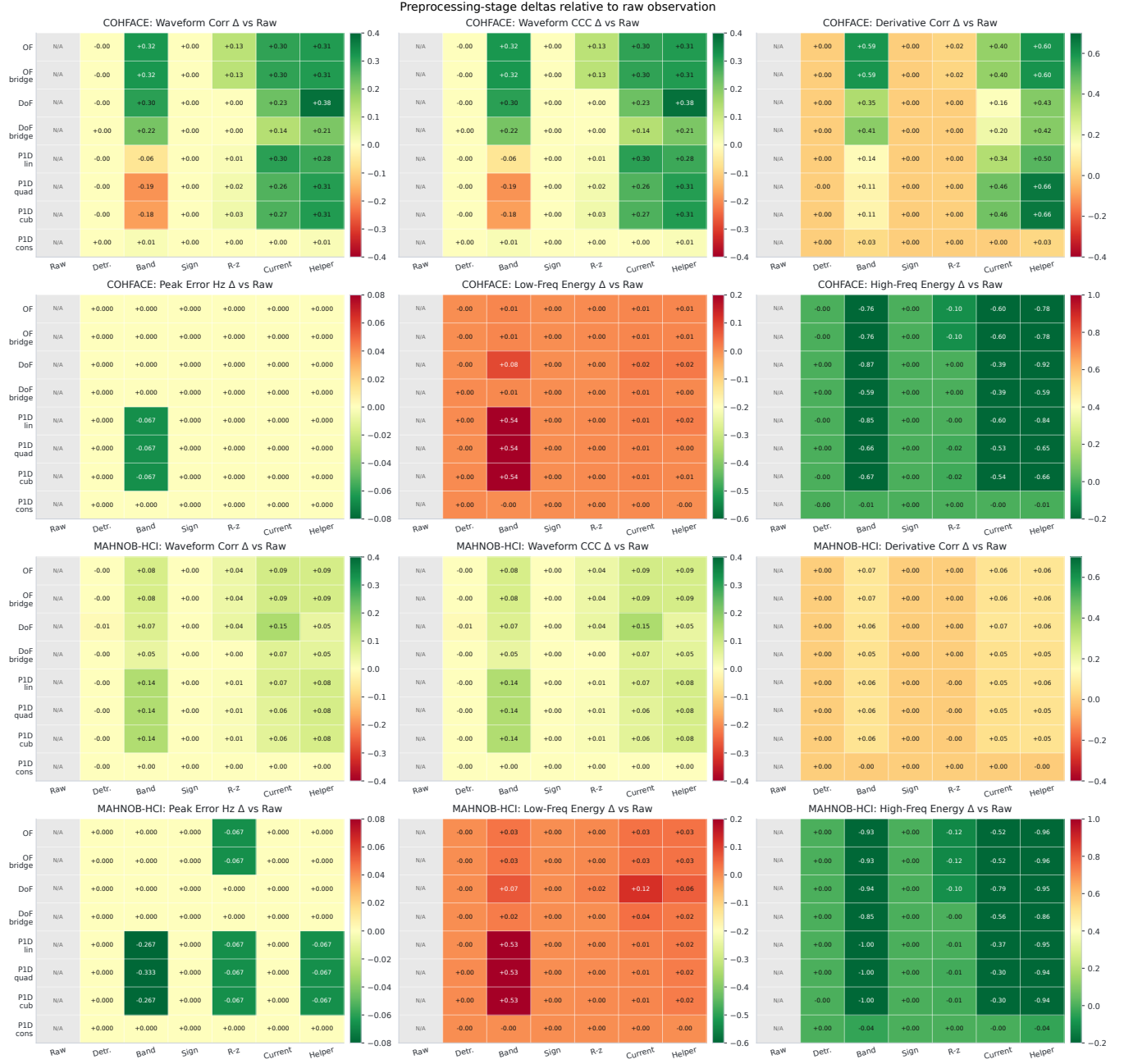
This figure marks the boundary between useful hard-regime diagnostics and the reported operating point. The fallback behaviour is informative about weak observability and nuisance, but it is not used to replace the predefined PARH-OSSM evaluation route.

This figure checks whether operating-point behaviour is stable within and across evaluation settings. Its role is to document readout-policy boundaries, not to introduce an additional tuned model variant.

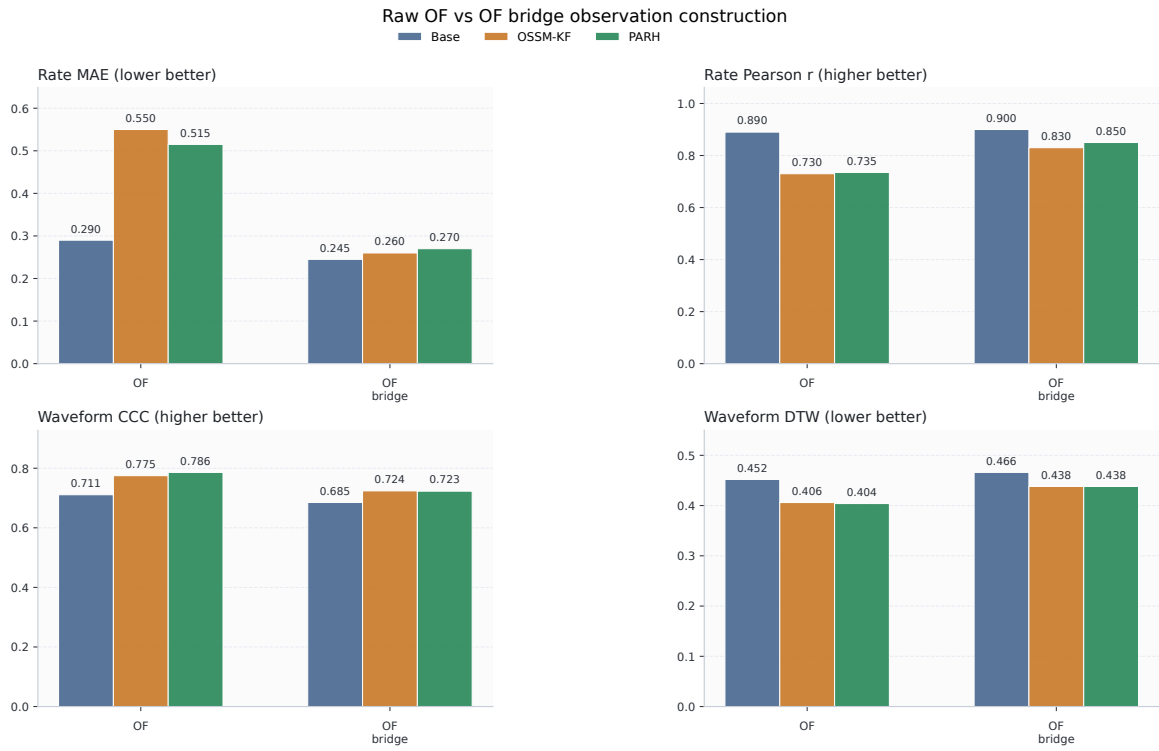
This overlay companion makes the OF construction claim visually auditable. The same reference grammar is used for raw OF and OF-bridge, so differences between the panels reflect observation construction rather than a change in plotting or case-selection logic.

This overlay companion performs the same construction-level check for DoF and DoF-bridge. It helps identify whether thresholded motion energy and signed centroid motion expose the same respiratory structure or fail through different nuisance mechanisms.

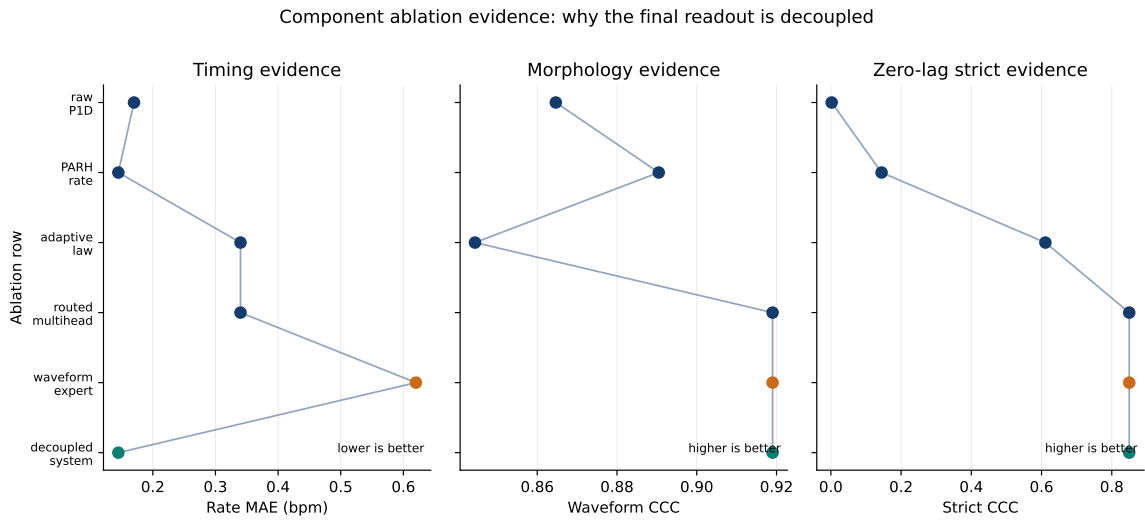
This atlas shows whether the clean-regime visual pattern in the main text is representative across observation classes. It supports the interpretation that COHFACE contains strong native respiratory evidence in several fixed



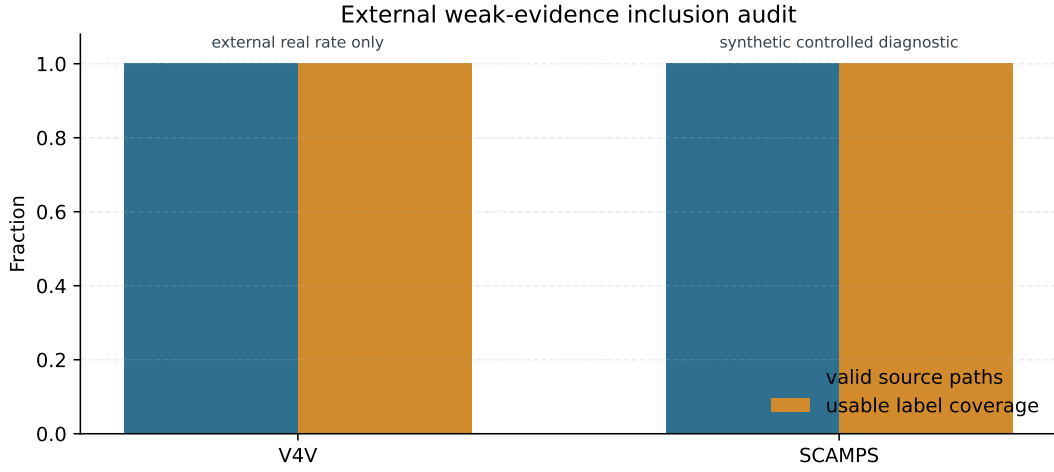
Supplementary Fig. S2. Preprocessing-stage delta heatmaps. The figure shows how raw, detrended, band-limited, sign-coded, robustly scaled, current-path, and helper views alter observation quality before state-space inference.



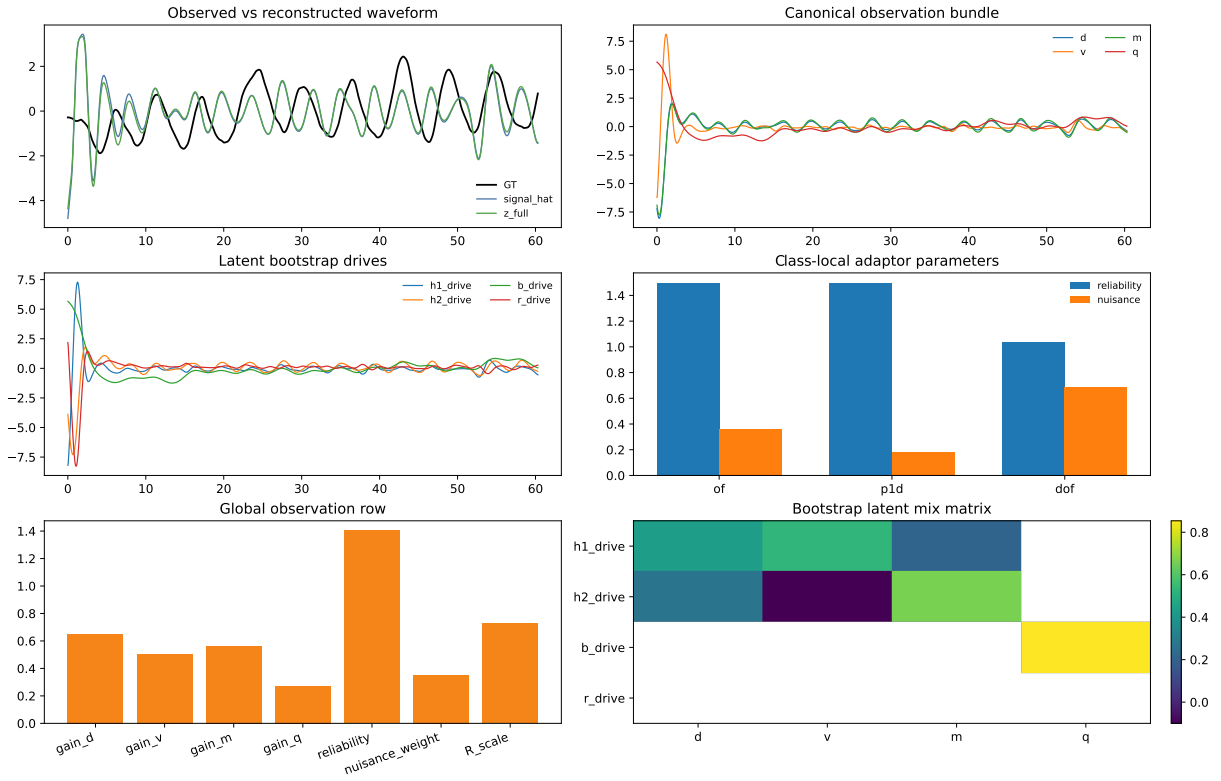
Supplementary Fig. S3. Optical-flow construction comparison. The raw optical-flow route and the displacement-compatible bridge are shown as distinct observation constructions rather than interchangeable preprocessing variants.



Supplementary Fig. S4. Component-ablation evidence summary. The figure separates performance-bearing architecture evidence from diagnostic or construction-only components.

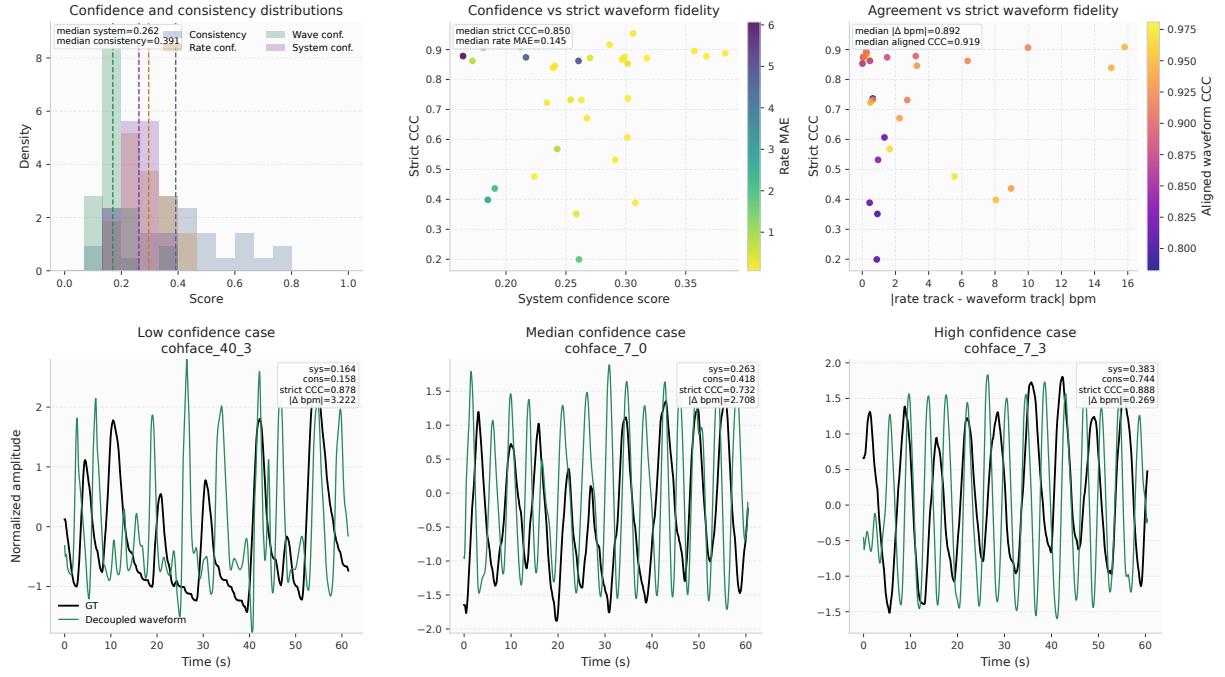


Supplementary Fig. S5. External weak-evidence summary. V4V and SCAMPS document rate-label and synthetic-control scope but are not pooled into the real waveform benchmark metrics.



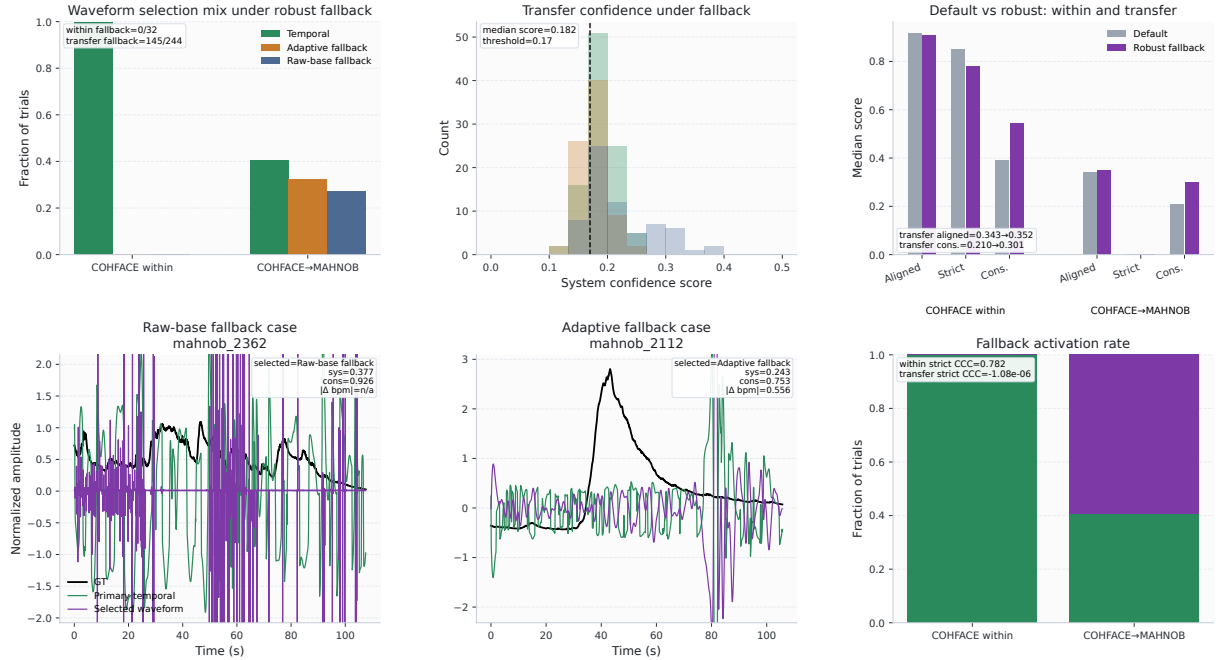
Supplementary Fig. S6. State-bundle and latent-bootstrap diagnostics. The figure visualizes the diagnostic observation summaries and latent state-drive traces retained to inspect the observation-to-state interface.

Decoupled system confidence, agreement, and case diagnostics



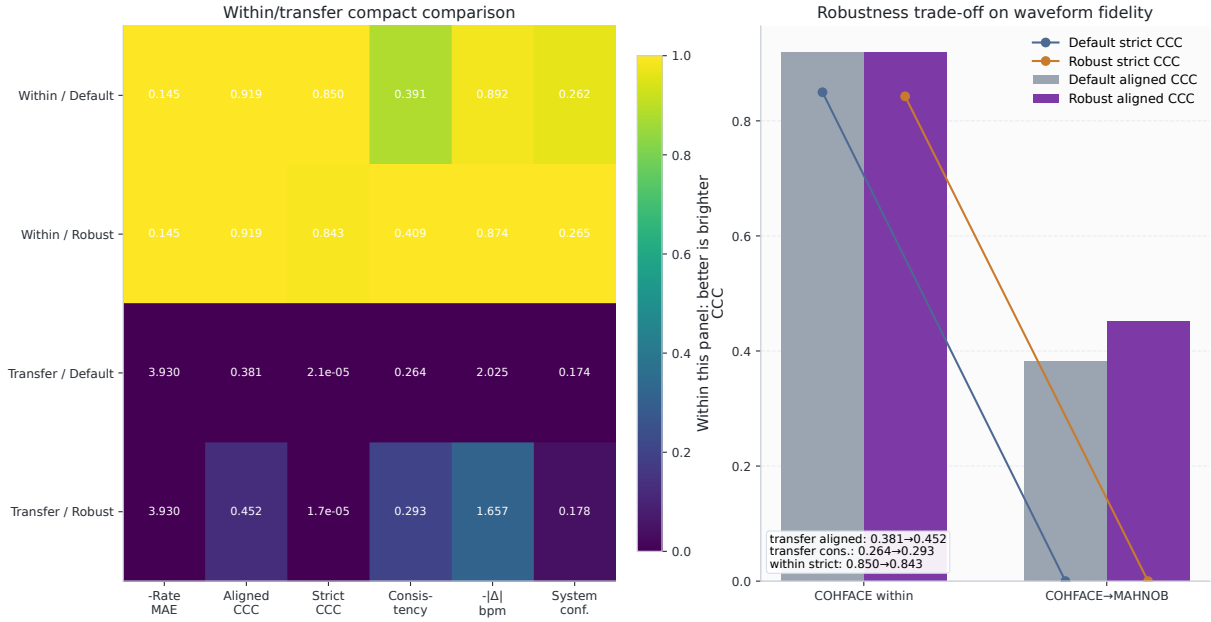
Supplementary Fig. S7. Decoupled-system confidence and rate-waveform agreement diagnostics. The figure summarizes why oscillatory and full-waveform readouts are interpreted separately.

Robust fallback diagnostics: selection, confidence, and hard-regime cases



Supplementary Fig. S8. Robust-fallback diagnostics. The fallback analyses are retained as hard-regime diagnostic evidence and are not promoted as the main operating point.

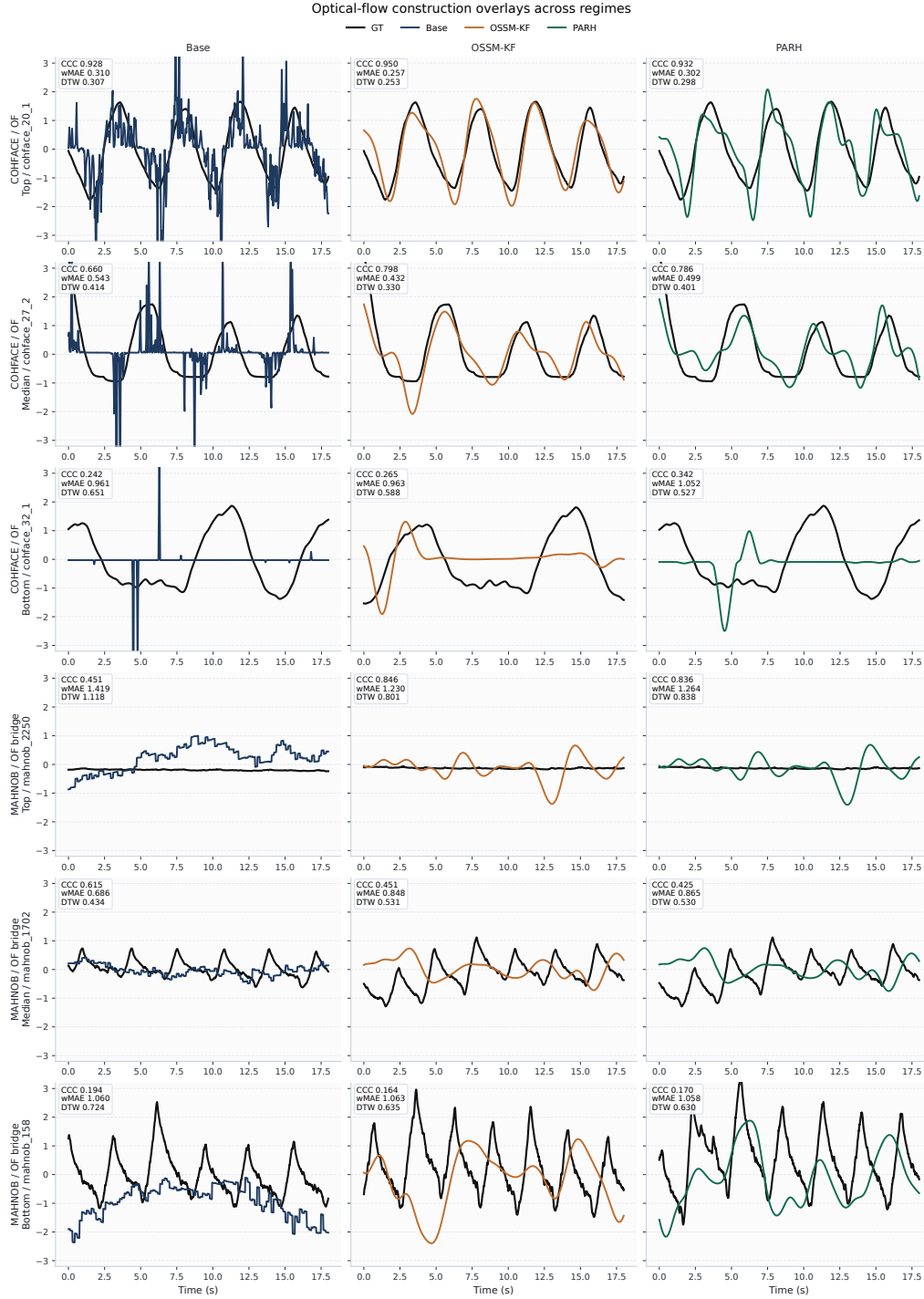
Compact within-vs-transfer view of default and robust decoupled systems



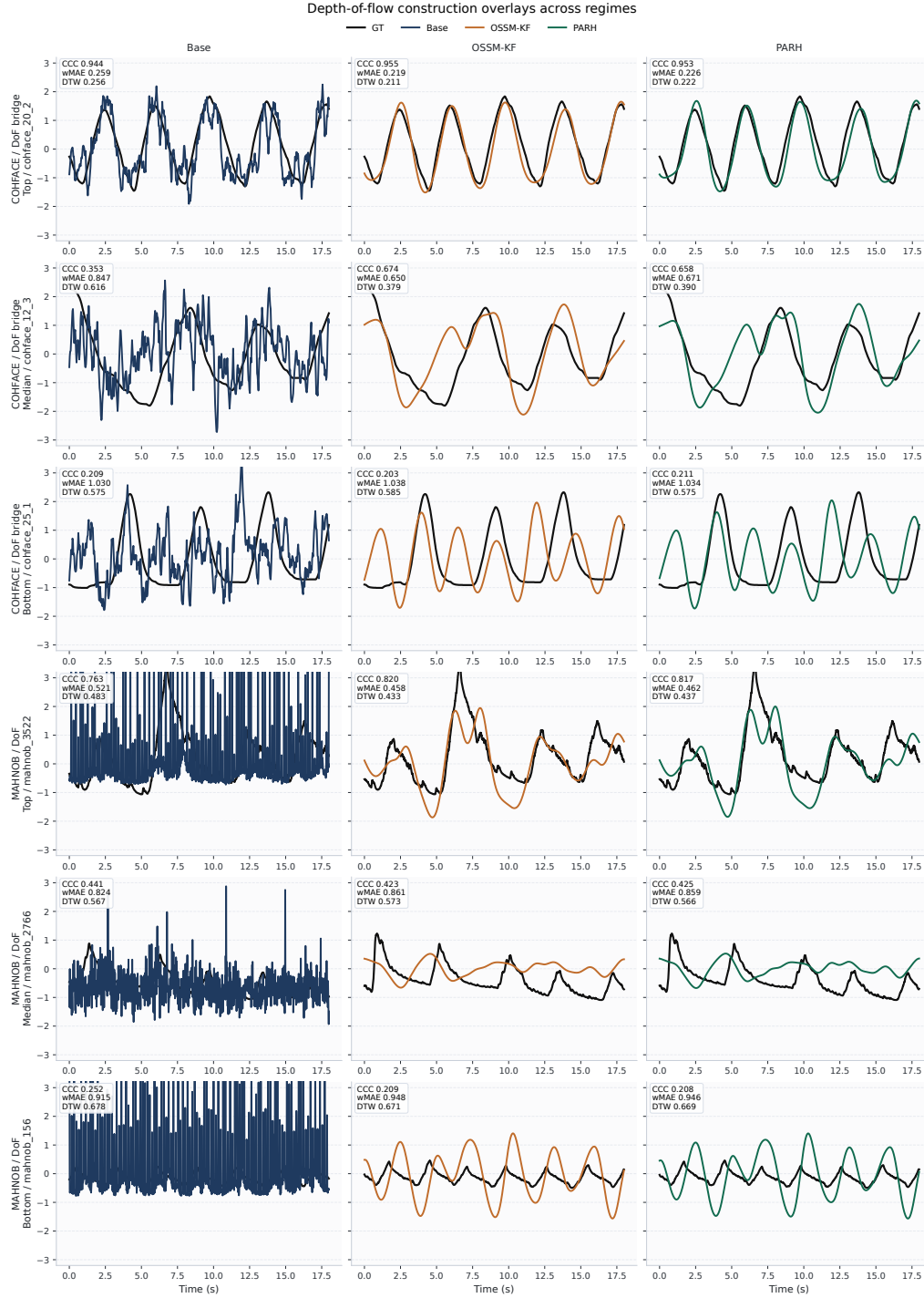
Supplementary Fig. S9. Within-versus-transfer operating-point comparison. The figure is retained as diagnostic context for readout policy boundaries.

observation routes.

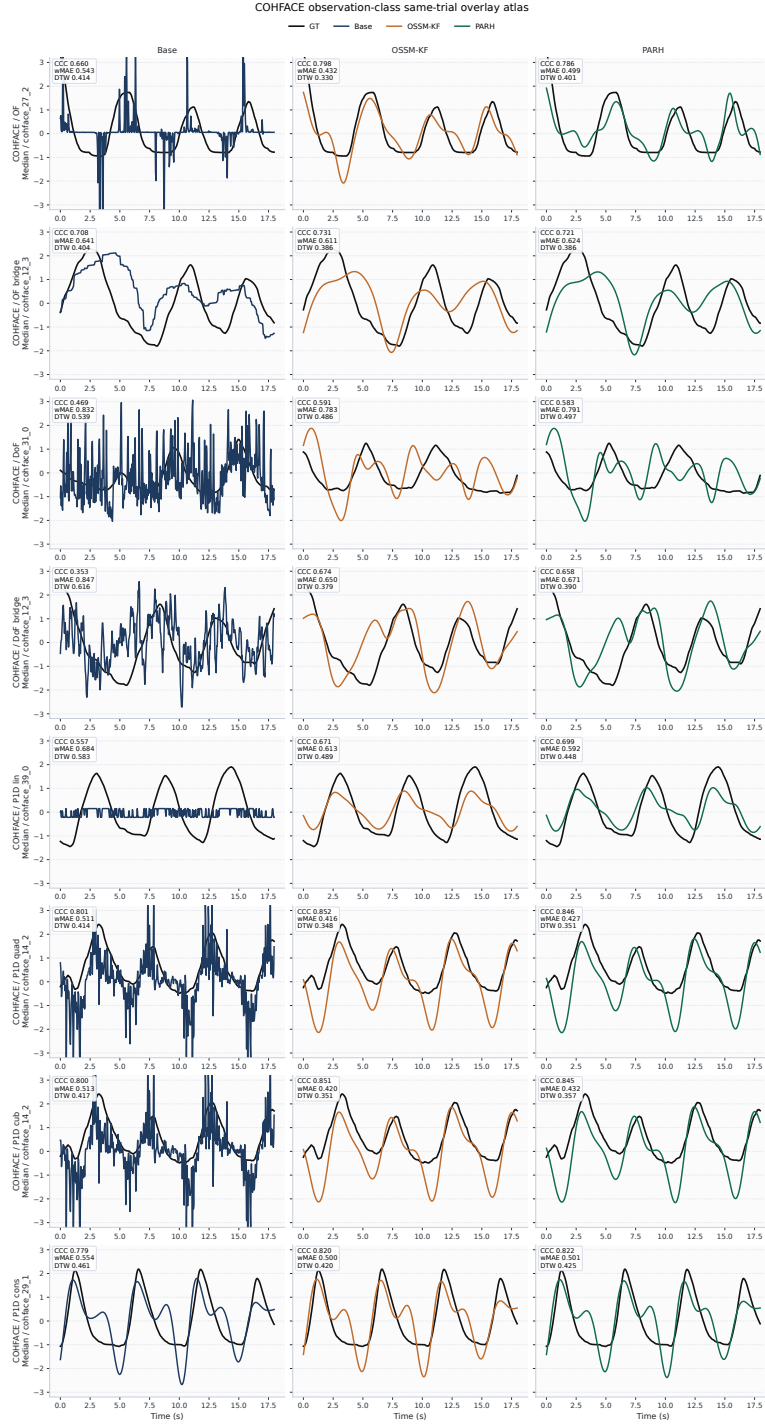
This atlas is the hard-regime counterpart to the COHFACE overlay companion. It shows that MAHNOB-HCI failures are not only final-readout errors; weak or inconsistent observation topology is already visible across the fixed observation bank.



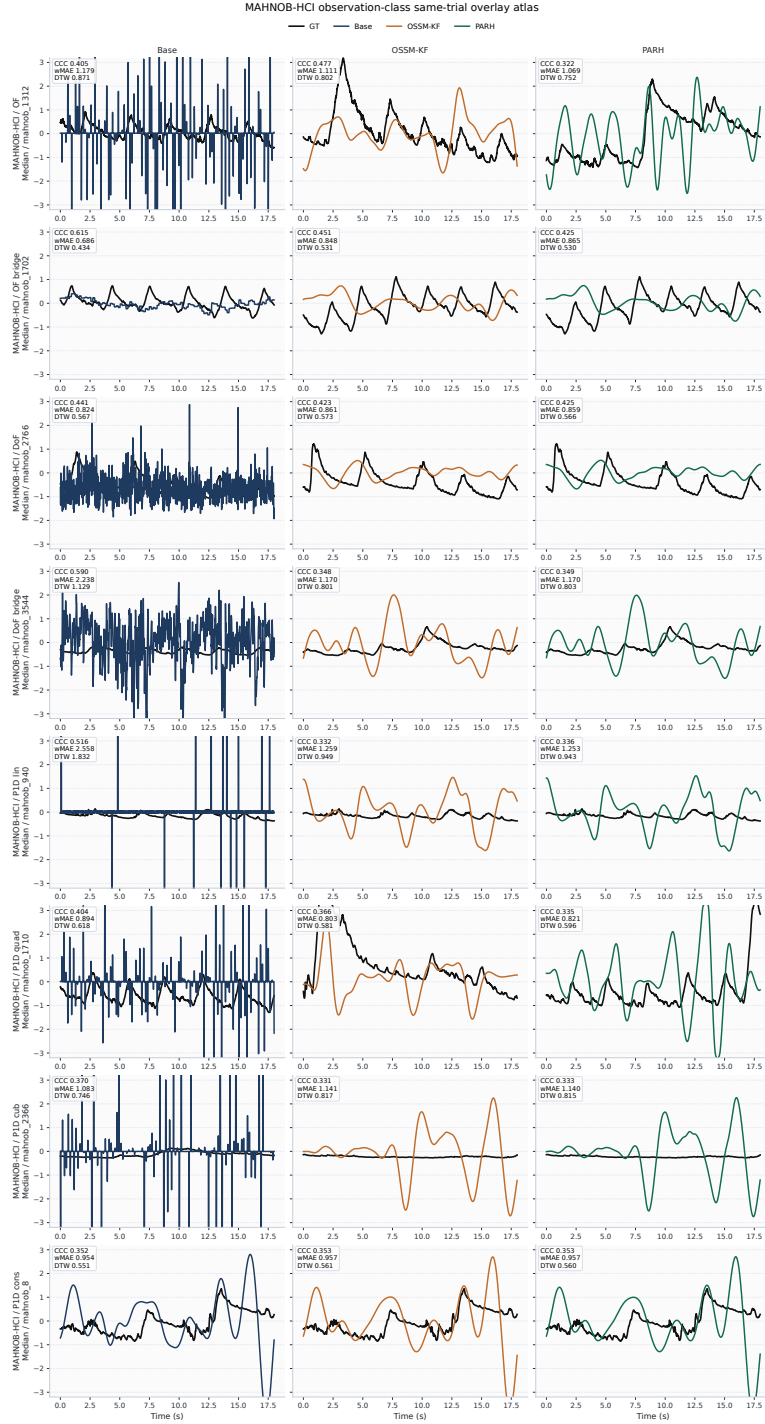
Supplementary Fig. S10. Optical-flow construction overlay companion. The panels show OF and OF-bridge behavior under the same visual grammar as the main overlay atlas.



Supplementary Fig. S11. Depth-of-flow construction overlay companion. The panels show DoF and DoF-bridge behavior as construction-level diagnostic context.



Supplementary Fig. S12. COHFACE observation-median all-class overlay atlas. The figure expands the clean-regime waveform topology evidence beyond the single main-text case.



Supplementary Fig. S13. MAHNOB-HCI observation-median all-class overlay atlas. The figure expands the hard-regime waveform topology evidence and helps localize low-observability behavior.